ACME METALS

Digital Material Passport

ID **DMP-2025-8842-1-002** - *Version* **1.0.0**
Issue Date **2025-03-18** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
Nordstahl Schmiedewerke GmbH Werkstraße 12 44135Dortmund DE info@nordstahl-schmiede.example.com	**ACME Steel Trading GmbH** Handelsweg 5c 76199Karlsruhe DE customer@example.com	*Order ID* **PO-2025-8842** - Quantity **19000 kg** *Delivery ID* **DN-2025-8842** - Quantity **3800 kg**

Product

Product Name	Forged round bars, rough turned	*Name (EN 10027-1)*	18CrNiMo7-6 +FP
Batch ID	BATCH-2025-8842	*Number (EN 10027-2)*	1.6587
Surface Condition	Rough turned	*Shape*	RoundBar - D 500 mm - Length 1500 mm
Delivery Condition	+FP - (Ferrite/pearlite annealed)	*Marking*	Head of bar stamped with grade, size, heat No., weight, forging No. Marking has to allow clear identification
Weight	9.5 t	*Packaging*	Bulk, max. weight of bar 13,5 mt
Diameter Tolerance	+3/-0 mm	*Coloring*	Yellow/black
Specification	8842-1/ GENERIC - FRT - Rev 1.0 - 2025-03-18	*Product Norm*	ISO 683-3
		Product Norm	ISO 6336-5

Heat Treatment

Process: **Annealing** - *Lot:* HT-LOT-2025-8842 - *Furnace:* F-12 - *Date:* 2025-03-10

Stage	*Temperature*	*Duration*	*Cooling*	*Atmosphere*
Soaking	650 C	6 h		Air

Chemical Analysis

Heat Number **HT-2025-4521** - *Melting Process* **EAF+LF+VD** - *Casting Method* **ContinuousCasting** - *Sample Location* **Ladle**

Symbol	*Unit*	*Min*	*Max*	*Actual*
C	%	0.15	0.21	0.18
Si	%	0.15	0.4	0.27
Mn	%	0.5	0.9	0.65
P	%		0.02	0.01
S	%		0.008	0.004
Cr	%	1.5	1.8	1.65
Ni	%	1.4	1.7	1.55
Mo	%	0.25	0.35	0.3
Cu	%		0.25	0.12
Ti	%		0.005	0.003

Symbol	*Unit*	*Min*	*Max*	*Actual*
Al	%	0.02	0.05	0.03
V	%			0.005
Sn	%		0.03	0.012
N	%	0.01	0.017	0.013
Ca	%		0.0025	0.0015
O	ppm		25	12
H	ppm		2	1.2
F1	%		0.5	0.24
F2	-	2	3.5	2.31

F1 = Cu+10*Sn: 0.24% **F2** = Al/N: 2.31-

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*
Tensile Strength[1] Specimen: 1/4T, L	Rm	1265 N/mm²	1200 N/mm²		
Yield Strength[1] Specimen: 1/4T, L	Rp0,2	905 N/mm²	850 N/mm²		
Elongation[1] Specimen: 1/4T, L	A5	13.5 %			
Reduction of area[1] Specimen: 1/4T, L	Z	52 %			
Impact test ISO-V +20°C on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling Specimen: 1/4T, L		58 / 62 / 60 J - Mean 60 - Min 58	40 J		
Impact test ISO-V -30°C on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling Specimen: 1/4T, L		45 / 48 / 47 J - Mean 46.7 - Min 45			
Surface hardness		185 HB	159 HB	207 HB	

Core hardness	178 HB	159 HB	207 HB

[1] on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling

Physical Properties

Property	*Actual*	*Target/Min*	*Maximum*	*Method*
Grain size	7	6		ISO 643
Purity A fine Plate II, inspected area approx. 200 mm²	2		3	ISO 4967 method B
Purity A thick Plate II, inspected area approx. 200 mm²	1.5		3	ISO 4967 method B
Purity B fine Plate II, inspected area approx. 200 mm²	1.5		2.5	ISO 4967 method B
Purity B thick Plate II, inspected area approx. 200 mm²	1		1.5	ISO 4967 method B
Purity C fine Plate II, inspected area approx. 200 mm²	1		2.5	ISO 4967 method B
Purity C thick Plate II, inspected area approx. 200 mm²	0.5		1.5	ISO 4967 method B
Purity D fine Plate II, inspected area approx. 200 mm²	1.5		2	ISO 4967 method B
Purity D thick Plate II, inspected area approx. 200 mm²	1		1.5	ISO 4967 method B
Purity DS Plate II, inspected area approx. 200 mm²	1		2	ISO 4967 method B
Purity K4	14		22	DIN 50602
Straightness	1.5 mm/m		2.5 mm/m	
Reduction ratio	6	4		
Residual magnetism	8 A/cm		15 A/cm	
Structure	Ferrite/pearlite			

✓ Fully formed structure ✓ Protected from reoxidation

Supplementary Tests

Test	*Result / limits*	*Method*
Ultrasonic test 100%	Yes - Pass	EN 10228-3 Type 1a class 3
Surface test 100% visually checked, clean surface free from defects	Yes - 100% visually checked, clean surface free from defects	
Radioactivity	Yes - max 100 Bq/kg	
Square cut obliquity max. 2% of diameter, no centerholes	Yes - Max. 2% of diameter, no centerholes	

✓ Free from mercury ✓ Free from cadmium ✓ 100% checked ✓ Sawn, free from burr

Hardenability – Jominy (ISO 642), values in HRC - +HH band; tested per ISO 642 or calculated values

Distance - from - quenched - end (mm)	**1.5**	**3**	**5**	**7**	**9**	**11**	**13**	**15**	**20**	**25**	**30**	**35**	**40**	**45**	**50**
Value [HRC]	46	46	45	44	44	44	42	42	40	39	38	37	37	30	30
Min	43	43	42	41	40	40	39	38	36	35	34	33	33		
Max	48	48	48	48	47	47	46	46	44	43	42	41	41		

Validation

We confirm that the goods described above comply with the requirements of the order and this inspection document was prepared in accordance with EN 10204:2004, type 3.1.

✓ 100% of the material is from Chinese sources

✓ Material is of non-Russian origin *(EU Regulation No. 833/2014)*

Validated by **Dr. Lena Krause**, Head of Quality Assurance, Quality Assurance - 2025-03-18